

HARARE, Zimbabwe

WMO: 677750

Lat: 17.930S

Lon: 31.094E

Elev: 1490

StdP: 84.66

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	7.0	-5.6	2.8	20.8	-3.1	3.5	20.5	10.2	15.1	9.0	15.0	2.2	80	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	11.3	31.2	16.3	30.2	16.2	29.2	16.3	20.6	25.0	20.1	24.6	19.7	24.0	4.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	17.1	21.0	19.0	16.6	20.9	18.5	16.1	20.8	66.7	25.2	65.0	23.8	63.5	24.4	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			1.9	22.0	1.3	1.1	1.0	22.8	0.2	23.4	-0.5	24.0	-1.5	24.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	21.7	21.8	21.2	19.4	17.0	14.7	14.5	16.9	20.1	21.9	22.4	21.8	(6)
(7)		DBStd	3.38	1.39	1.33	1.45	1.85	1.83	1.89	1.86	2.31	2.52	2.47	2.16	1.49	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	361	1	0	1	11	48	108	119	57	11	4	2	1	(9)
(10)		CDD10.0	3444	363	329	347	281	218	142	141	215	303	369	372	364	(10)
(11)		CDD18.3	764	106	96	89	42	8	1	1	13	64	114	124	107	(11)
(12)		CDH23.3	5373	487	501	456	279	99	25	27	203	697	1102	959	537	(12)
(13)		CDH26.7	1226	77	79	61	26	4	0	1	23	174	369	304	108	(13)
(14)	Wind	WSAvg	3.3	3.1	3.0	3.1	3.0	2.7	2.8	3.3	3.5	4.1	4.2	3.9	3.2	(14)
(15)	Precipitation	PrecAvg	784	197	142	84	39	8	3	2	8	37	92	163		(15)
(16)		PrecMax	1272	463	420	249	116	67	20	35	27	70	104	293	382	(16)
(17)		PrecMin	418	1	0	1	0	0	0	0	0	0	2	3		(17)
(18)		PrecStd	191	110	91	62	35	14	5	6	16	30	62	95		(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	30.0	29.8	28.9	27.1	25.8	25.9	28.8	31.8	33.2	32.7	31.0	(19)
(20)			MCWB	18.0	18.4	18.3	16.9	14.3	12.8	12.0	13.1	15.3	16.2	16.7	16.9	(20)
(21)		2%	DB	28.5	28.6	28.1	27.2	25.7	23.9	24.0	27.1	30.0	31.4	31.1	29.1	(21)
(22)			MCWB	18.4	18.0	18.2	16.5	14.0	12.6	11.5	13.0	14.8	15.9	16.7	17.4	(22)
(23)		5%	DB	27.2	27.5	27.1	26.1	24.2	22.4	22.6	25.8	28.8	30.1	29.9	27.8	(23)
(24)			MCWB	18.6	18.2	18.3	16.3	13.7	12.4	11.3	12.7	14.4	15.7	16.6	17.8	(24)
(25)		10%	DB	26.1	26.3	26.0	25.0	23.1	21.1	21.0	24.0	27.2	28.9	28.6	26.2	(25)
(26)			MCWB	18.8	18.3	18.1	16.0	13.5	11.9	11.2	12.2	14.1	15.5	16.6	18.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	21.0	21.0	19.5	16.7	15.1	14.2	15.3	17.2	18.9	19.9	20.7	(27)
(28)			MCDWB	25.3	25.6	25.1	24.5	21.7	21.0	21.9	24.2	26.1	28.1	26.1	25.1	(28)
(29)		2%	WB	20.4	20.2	20.1	18.6	15.7	14.2	13.1	14.4	16.2	18.0	19.2	20.1	(29)
(30)			MCDWB	24.7	24.7	24.7	24.0	21.9	21.2	20.9	23.7	26.4	26.9	24.9	24.7	(30)
(31)		5%	WB	20.0	19.8	19.6	18.0	15.1	13.3	12.4	13.6	15.6	17.3	18.6	19.7	(31)
(32)			MCDWB	24.2	24.1	24.0	23.0	21.6	20.1	19.8	22.9	26.0	26.2	24.4	24.1	(32)
(33)		10%	WB	19.5	19.4	19.1	17.3	14.5	12.6	11.7	13.0	15.0	16.7	18.1	19.2	(33)
(34)			MCDWB	23.5	23.5	23.2	22.0	20.9	19.0	18.8	21.9	25.1	25.4	24.0	23.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	10.2	11.5	12.6	12.9	13.0	14.0	14.4	13.4	11.3	9.6	(35)
(36)			MCDBR	11.2	11.6	11.7	12.6	14.0	14.6	15.4	15.6	15.1	14.7	13.4	12.0	(36)
(37)			MCWBR	3.5	3.8	3.9	4.4	5.3	5.7	6.0	5.4	5.0	4.6	4.2	3.9	(37)
(38)		5% WB	MCDWB	8.9	9.5	9.7	10.7	11.9	13.0	12.8	13.6	13.6	12.7	10.5	9.4	(38)
(39)			MCWBR	3.4	3.6	3.8	4.1	5.0	5.6	5.5	5.4	5.0	4.5	3.8	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.336	0.323	0.289	0.270	0.265	0.276	0.317	0.389	0.385	0.367	0.351	(40)	
(41)		taud	2.548	2.550	2.593	2.683	2.686	2.656	2.572	2.400	2.220	2.259	2.377	2.491	(41)	
(42)		Ebn at Noon	1006	997	987	988	975	962	956	942	907	937	970	992	(42)	
(43)		Edh at Noon	110	109	101	86	79	79	88	112	144	144	130	117	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.07	6.11	5.74	5.47	5.10	4.61	4.75	5.71	6.41	6.75	6.30	5.97	(44)	
(45)		RadStd	0.46	0.55	0.45	0.34	0.26	0.26	0.27	0.23	0.22	0.36	0.41	0.48	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page